



41-MAVZU · V BOB. DRF VA DEPLOY

Django loyihasini serverga joylash (Deploy)

Deploy nima, development va production farqi, hosting turlari; loyihani tayyorlash: DEBUG, SECRET_KEY, ALLOWED_HOSTS, gunicorn, whitenoise, dj-database-url, requirements.txt, collectstatic; Render'ga bosqichma-bosqich joylash, PostgreSQL, superuser; PythonAnywhere muqobili; deploy'dan keyingi tekshiruv.

8-sinf · Web Full-stack dasturlash

Bugun nimani o'rganamiz?

- **Deploy** nima: "kompyuterimda ishlaydi" → "internetda ishlaydi"
- **Development** va **production** farqi
- Loyihani serverga **tayyorlash**: xavfsizlik, statik fayllar, server dasturi
- **Render**'ga PostgreSQL bilan joylash (30-loyiha)
- Joylangan saytni tekshirish va yangilash

Deploy nima?

- Hozircha blog faqat `127.0.0.1:8000` da — uni hech kim ko'ra olmaydi

Deploy — dasturni internetga ulangan serverga o'rnatish va ishga tushirish



Uy oshxonasi va restoran: uyda pishirish (development) — xato qilsangiz hech kim bilmaydi. Restoran (production) — sanitariya qoidalari (xavfsizlik), katta oshxona (server), 24/7 ish, ko'p mijoz. Retsept bir xil — talablar boshqa.

Development va production

	Development	Production
Qayerda	Sizning kompyuteringiz	Internetdagi server
<code>DEBUG</code>	<code>True</code> — batafsil xatolar	<code>False</code> — xatolar yashirin!
Server	<code>runserver</code>	gunicorn
Statik fayllar	Django o'zi	<code>collectstatic</code> + WhiteNoise
Maxfiy kalitlar	<code>.env</code> fayl	Muhit o'zgaruvchilari
Manzil	<code>127.0.0.1:8000</code>	<code>https://maktab-blog.onrender.com</code>

Hosting turlari

Platforma	Turi	Afzalligi	E'tibor bering
Render	PaaS	GitHub'dan avtomatik, PostgreSQL, HTTPS	Bepul tarifda "uxlaydi"
Railway	PaaS	Juda oson	Bepul limit cheklangan
PythonAnywhere	Python hosting	O'quvchilarga eng oddiy	Bepulda PostgreSQL yo'q
VPS	Virtual server	To'liq nazorat	Hammasini o'zingiz sozlaysiz

- **PaaS** — server va tarmoqni platforma boshqaradi, siz faqat kod berasiz



Tariflar o'zgaradi — joylashdan oldin platforma saytida tekshiring

BO'LIM

1-bosqich

Loyihani production uchun tayyorlash

Kerakli paketlar

Terminal

```
pip install gunicorn whitenoise dj-database-url  
pip freeze > requirements.txt
```

Paket	Vazifasi
<code>gunicorn</code>	Production server (<code>runserver</code> o'rniga)
<code>whitenoise</code>	Statik fayllarni (CSS, JS) tez va xavfsiz berish
<code>dj-database-url</code>	Bazani bitta <code>DATABASE_URL</code> satridan sozlash

- `gunicorn` Windows'da ishlamaydi — normal: u faqat serverda (Linux).
Kompyuteringizda `runserver`

settings.py — xavfsizlik

Python

```
import os
import dj_database_url

SECRET_KEY = os.environ["SECRET_KEY"]          # yo'q bo'lsa — xato (ataylab!)
DEBUG = os.getenv("DEBUG", "False") = "True"   # standart — False
ALLOWED_HOSTS = os.getenv("ALLOWED_HOSTS", "127.0.0.1,localhost").split(",")
CSRF_TRUSTED_ORIGINS = [f"https://{h}" for h in ALLOWED_HOSTS
                        if h not in ("127.0.0.1", "localhost")]

if not DEBUG:
    SECURE_PROXY_SSL_HEADER = ("HTTP_X_FORWARDED_PROTO", "https")
    SESSION_COOKIE_SECURE = True
    CSRF_COOKIE_SECURE = True
```

- Lokal `.env` ga `DEBUG=True` qo'shing — kompyuterda qulay ishlash uchun

settings.py — baza va statik fayllar

Python

```
MIDDLEWARE = [  
    "corsheaders.middleware.CorsMiddleware",  
    "django.middleware.security.SecurityMiddleware",  
    "whitenoise.middleware.WhiteNoiseMiddleware",      # ← Security'dan keyin  
    # ...  
]  
  
DATABASES = {  
    "default": dj_database_url.config(  
        default=f"postgres://{os.getenv('DB_USER')}:{os.getenv('DB_PASSWORD')}@"  
            f"{os.getenv('DB_HOST', 'localhost')}:5432/{os.getenv('DB_NAME')}",  
        conn_max_age=600,  
    )  
}  
  
STATIC_URL = "static/"  
STATIC_ROOT = BASE_DIR / "staticfiles"  
STORAGES = {  
    "default": {"BACKEND": "django.core.files.storage.FileSystemStorage"},  
    "staticfiles": {"BACKEND": "whitenoise.storage.CompressedManifestStaticFilesStorage"},  
}
```

Production'ning oltin qoidalar

- 1 **DEBUG = False** — aks holda xato sahifasi buzg'unchiga kod, sozlamalar va hatto parollarni ko'rsatadi
- 2 **SECRET_KEY** **maxfiy va yangi** — GitHub'dagi yoki darsdagisini ishlatmang
- 3 **ALLOWED_HOSTS** — faqat sizning domeningiz

Yangi SECRET_KEY yaratish

```
python -c "from django.core.management.utils import get_random_secret_key; print(get_random_secret_key())"
```

Build skripti va tekshiruv

build.sh

```
#!/usr/bin/env bash
set -o errexit # xato bo'lsa - to'xtash
pip install -r requirements.txt
python manage.py collectstatic --no-input
python manage.py migrate
```

Terminal

```
git update-index --chmod=+x build.sh # Windows: bajariladigan qilish (macOS/Linux: chmod +x)
python manage.py check --deploy # production maslahatlari
python manage.py collectstatic # staticfiles/ (.gitignore ga!)
git add . && git commit -m "Production uchun tayyorlandi" && git push
```

BO'LIM

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Render'ga joylash

Render: baza va Web Service

- 1 **render.com** — GitHub akkaunti bilan kiring
- 2 **New + → Postgres** → `maktab-blog-db`, region **Frankfurt** → **Internal Database URL** ni nusxalang
- 3 **New + → Web Service** → repozitoriyangiz

Maydon	Qiymat
Name	<code>maktab-blog</code>
Runtime	Python 3
Build Command	<code>./build.sh</code>
Start Command	<code>gunicorn config.wsgi:application</code>
Instance Type	Free

Muhit o'zgaruvchilari

Key	Value
SECRET_KEY	Yangi uzun tasodifiy kalit
DEBUG	False
ALLOWED_HOSTS	maktab-blog.onrender.com
DATABASE_URL	Internal Database URL
PYTHON_VERSION	Masalan 3.12.7

- 4 **Create Web Service** → **Logs**: install → collectstatic → migrate → gunicorn
- 5 "Your service is live 🎉"
- 6 Admin: **Shell** → `python manage.py createsuperuser`

Avtomatik deploy (CI/CD)

Matn

 git push →  GitHub — avtomatik →  Render: build.sh → gunicorn ⇌  Render PostgreSQL

Har bir git push → sayt avtomatik qayta joylanadi

Terminal

```
git add .  
git commit -m "Izohlar funksiyasi qo'shildi"  
git push          # yangi migratsiyalar build.sh'da bajariladi
```

Tekshiruv ro'yxati

Tekshiruv	✓
Bosh sahifa ochiladi, CSS ishlayapti	☐
<code>/admin/</code> ga kirish, maqola qo'shish	☐
Ro'yxatdan o'tish, CRUD ishlaydi	☐
<code>/maqola/9999/</code> — chiroyli 404 (sariq xato EMAS!)	☐
<code>/api/maqolalar/</code> JSON qaytaradi	☐
Manzilda  HTTPS	☐
Telefonda to'g'ri ko'rinadi	☐
JS front-end (Netlify) Render API bilan ishlaydi (CORS)	☐

Xatolarni topish — Logs

Belgi	Sabab
<code>DisallowedHost</code>	Domen <code>ALLOWED_HOSTS</code> da yo'q
CSS yo'q (oddiy matn)	WhiteNoise yoki <code>collectstatic</code> muammosi
<code>CSRF verification failed</code>	<code>CSRF_TRUSTED_ORIGINS</code> da <code>https://</code> domen yo'q
<code>ModuleNotFoundError</code>	Paket <code>requirements.txt</code> da yo'q
"Build failed"	Build log'ini o'qing

⚠ Muammoni "tuzatish" uchun serverda **hech qachon** `DEBUG=True` qo'ymang!

- Media fayllar bulutda vaqtinchalik — haqiqiy loyihada Cloudinary / S3

Muqobil: PythonAnywhere

- 1 **pythonanywhere.com** — bepul "Beginner" akkaunt
- 2 **Bash:** `git clone ...`, virtual muhit, `pip install -r requirements.txt`
- 3 **Web → Add a new web app → Manual configuration;** WSGI faylida
`DJANGO_SETTINGS_MODULE = "config.settings"`
- 4 Bepulda PostgreSQL yo'q → `sqlite:///db.sqlite3`; `/static/` → `staticfiles`
- 5 **Reload** → `USERNAME.pythonanywhere.com`

Amaliyot (30-loyiha)

- 1 Loyihani tayyorlang; `check --deploy` ogohlantirishlarini o'qing
- 2 Render'da PostgreSQL + Web Service → joylang
- 3 **Tekshiruv ro'yxatini** to'liq bajaring
- 4 Footer matnini o'zgartiring → push → avtomatik qayta joylanishni kuzating
- 5 `README.md`: sayt havolasi, skrinshotlar, "Qanday ishga tushirish"

Savol

Production serverda **DEBUG** qanday bo'lishi kerak va nima uchun?

- A** True — xatolarni ko'rish uchun
- B** False — xato sahifalari maxfiy ma'lumotlarni oshkor qilmasligi uchun
- C** Ahamiyati yo'q
- D** None

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

Production'da **runserver** o'rniga ishlatiladigan server dasturi?

A whitenoise

B pip

C psql

D gunicorn

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

Saytingiz `DisallowedHost` xatosini berdi. Sababi?

A Baza ulanmagan

B CSS topilmadi

C Domen `ALLOWED_HOSTS` da yo'q

D Parol noto'g'ri

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

Render'ga ulangan loyihani yangilash uchun nima qilinadi?

- A** Serverga qayta ro'yxatdan o'tish
- B** Loyihani o'chirib qayta yaratish
- C** O'zgarishlarni GitHub'ga `git push` qilish
- D** Yangilab bo'lmaydi

Javobni tanlang — to'g'ri variant yashil bo'ladi



Uyga vazifa

- 1 IV bob yakunlovchi loyihangizni ham internetga joylang
- 2 Barcha havolalarni **GitHub profili README**'siga va **portfolio** saytingizga qo'shing — bu sizning birinchi **"rezyume"**ingiz!

Yodda tuting

- Deploy — internetdagi serverga joylash; production ≠ development
- `DEBUG=False`, maxfiy `SECRET_KEY`, `ALLOWED_HOSTS`, `CSRF_TRUSTED_ORIGINS`
- `gunicorn`, `whitenoise`, `django-database-url`; `requirements.txt`
- `build.sh`: install → collectstatic → migrate; start: `gunicorn config.wsgi:application`
- Render: Postgres + Web Service + muhit o'zgaruvchilari → har push'da deploy



Savollar bormi?

Keyingi mavzu: **42. Yakuniy nazorat ishi — IV–V boblar (Django, DRF, Deploy) va yakuniy loyiha**

Batafsil matn, misollar va testlar: **41-django-deploy.html** darsida